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Artificial Intelligence Agents for Personalized Adaptive Learning

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Abstract

This paper presents AIA-PAL (Artificial Intelligence Agents for Personalized Adaptive Learning), a framework designed to improve student outcomes through advanced Human-Agent Interaction (HAI). It addresses limitations in current intelligent tutoring systems by enabling dynamic scaffolding and personalized learning paths. AIA-PAL uses LangGraph for decision-making and a multi-agent system via CrewAI to adapt learning in real time. Its dual-layer approach integrates real-time monitoring with specialized agents (tutor, practical, and teacher) to provide targeted pedagogical support. Large language models are employed to enable personalized dialogue, improving responsiveness to queries. To ensure pedagogical accuracy and minimize hallucinations, AIA-PAL uses retrieval-augmented generation grounded in teacher-provided content and validated through structured human oversight. Preliminary results indicate improved scaffolding and personalized learning, boosting engagement and efficiency. The paper details the framework's architecture, methods, and outcomes, highlighting real-time adaptability and HAI as key to effective learning.

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Keywords:
Large Language Models; Multi-Agent System; Personalized Adaptive Learning; Intelligent Tutoring Systems; LangGraph; CrewAI; Human-Agent Interaction; Retrieval-Augmented Generation.

1. Introduction

Artificial Intelligence (AI) has significantly reshaped Personalized Learning (PL), with adaptive learning technologies playing a central role [1, 2, 3]. These systems tailor educational content and interactions to individual student performance and pace [4, 5, 6]. Traditional methods often lack such personalization, prompting the rise of Intelligent Tutoring Systems (ITS), which improve learning through advanced Human-Agent Interaction (HAI). It is increasingly recognized that fostering effective communication and understanding between humans and AI systems is vital for their successful integration into various applications [24, 28, 32]. Conversational agents, as shown by [7], enable interactive dialogue, immediate feedback, and adaptive support using Natural Language Processing and Machine Learning. ITS

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can achieve outcomes comparable to human tutors [8], promoting deep learning via real-time performance analysis and personalized feedback. Despite their promise, ITS often lack effective scaffolding—the gradual adjustment of support as learners gain competence. Many systems fail to detect subtle performance cues, leading to overreliance on hints and feedback rather than fostering independent problem-solving. AI-powered chatbots also offer benefits such as homework assistance and personalized feedback [9], but concerns remain over their potential impact on critical thinking through excessive dependence.

Intelligent agents [10, 11] are designed to assess student responses, track progress, and adapt instructional strategies to individual needs. Recent work [12, 13] highlights their potential to enhance learning through personalization and adaptivity. Beyond student interaction, AI agents automate tasks such as grading, attendance, and curriculum planning. However, overreliance on automation risks diminishing human oversight, introducing errors in complex decisions, and depersonalizing educational administration, potentially undermining student support and engagement. As shown by [14], AI also influences communication and support in online learning, raising concerns over student privacy, discomfort from surveillance, and blurred boundaries where human empathy is essential. Bias in AI systems, stemming from their reliance on large datasets, may perpetuate existing inequalities and stereotypes. Additionally, automating core educational functions risks marginalizing teachers, reducing opportunities for meaningful student interaction. These challenges underscore the need for balanced AI integration that prioritizes human oversight, ethical standards, and learner development.

This paper addresses the aforementioned limitations through the development of AIA-PAL (Artificial Intelligence Agents for Personalized Adaptive Learning) framework, where we build a Multi-Agent System (MAS) around LangGraph as the central decision-making engine. The framework includes specialized agents (Student Profile, Monitor, and Decision Engine) alongside educational agents implemented via CrewAI (Tutor, Practical, and Teacher Profile). Each agent is assigned distinct responsibilities, working collaboratively to deliver personalized, adaptive, and engaging learning experiences. The system is grounded in a defined pedagogical framework, using teacher-provided instructional materials as the foundation for its knowledge base. To maintain content accuracy and minimize hallucinations, AIA-PAL integrates a Retrieval-Augmented Generation (RAG) pipeline, enabling contextually grounded educational dialogues powered by advanced Large Language Models (LLMs) such as GPT-4. Teachers are incorporated as validators within the MAS, ensuring the pedagogical integrity and alignment of generated materials with educational standards. Emphasizing the importance of HAI, this comprehensive approach ensures continuous real-time adjustments to content and PL pathways, including scaffolding. The following sections examine the current state of AI in education and related work in adaptive learning systems, the approach used in developing the AI agents, and the system's architecture, followed by a practical implementation using the LangGraph and CrewAI framework. We then present the initial findings observed after conducting a pilot test. Finally, we conclude by discussing potential future perspectives and areas of improvement.

2. Current state of AI in education

2.1. Adaptive Learning Systems

Notable adaptive learning systems include Khanmigo [15] and Duolingo [16]. Developed by Khan Academy, Khanmigo serves as an AI tutor and assistant designed to foster critical thinking and problem-solving across subjects such as mathematics, humanities, and coding, targeting students from elementary to college levels. Integrated with Khan Academy's content library, it supports educational standards and enables teacher and parent oversight. However, it lacks the option for users to upload custom course content, limiting opportunities for personalized curricula. Enhancements such as allowing course customization, enabling real-time interactive feedback during exercises, and refining adaptive learning algorithms to accommodate diverse learning styles and paces could further improve its educational utility and responsiveness to individual learner needs.

Duolingo [16] is an online language learning platform that offers gamified lessons and activities to make language acquisition accessible and engaging. It adapts to individual learning speeds through a user-friendly interface and features interactive exercises that reinforce vocabulary and grammar, along with motivational tools such as a streak counter to encourage consistent practice. However, the platform has limitations, including a lack of advanced content for higher-level learners and simplistic translations that may overlook contextual and idiomatic nuances. To address

these issues, Duolingo could incorporate more complex lessons focused on advanced language structures and real-world communication. Additionally, integrating AI-driven personalized feedback, particularly for speech recognition and pronunciation, could enhance the platform's responsiveness to diverse learning needs and styles.

2.2. AI-Driven Adaptive Learning and Scaffolding in ITS

Studies [17, 18, 19, 20] show that AI enhances student performance by providing timely feedback, improving engagement, retention, teaching methods, and administrative efficiency. Related work highlights AI's role in creating sustainable education systems by enhancing accessibility and quality through technology-enabled learning [21, 22]. However, traditional ITS [37] face limitations, including difficulty replicating human-like feedback and interaction, adapting to students' motivational states, and biases stemming from limited training data. In educational terms, "to scaffold" refers to a supportive teaching approach that gradually guides students toward stronger understanding and greater independence. This method provides a temporary framework of guidance, adjusting support based on student performance. As understanding improves, the support is reduced to foster autonomy. Scaffolding is essential for promoting deeper engagement and mastery of complex concepts. Research on scaffolding [23, 25] shows its efficacy across various learning environments, especially online. It enhances learning by offering tailored support to help students grasp concepts they might not master alone. This aligns with Vygotsky's Zone of Proximal Development, which suggests that learners can accomplish more with capable guidance. A meta-analysis [25] on online learning confirmed that scaffolding significantly improves outcomes, though effectiveness varies by discipline, learning goals, and types of scaffolding. This suggests that while generally beneficial, scaffolding must be carefully tailored to different educational contexts. However, most current systems use static granularity, meaning the level of support does not evolve with the learner's proficiency. This limitation prevents the system from adapting to the individual needs of learners.

The ITS utilize rule-based tutoring [26], where a cognitive model representing the necessary skills and knowledge for a student is framed using production rules. These rules are conditional statements (if-then) that dictate responses to various learning scenarios according to the student's actions or answers. Each rule directly relates to particular student skills, designed to customize the learning journey. As students engage with these systems, the rules assist in providing tailored feedback and guidance, adjusting the teaching method based on the student's ongoing performance. Notable rule-based tutoring systems that have been widely adopted include Cognitive Tutor/MATHia [27], OATutor [29], ASSISTments [30], and Apprentice Tutor [31]. The prevailing method for representing skills in ITS is predominantly rule-based. Skills are defined as discrete and triggered under certain conditions. However, this method does not adequately capture the compositional and hierarchical aspects of skill acquisition and application. According to [33], the concept of adaptive learning emphasizes real-time adaptability in teaching strategies to cater to individual student characteristics, performance, and personal development. Adaptive learning systems actively adjust instructional strategies based on monitored changes in learners. This adaptability involves data-driven decision-making, which is crucial for tailoring educational experiences to individual needs effectively. The goal is to optimize learning outcomes by dynamically adjusting the level of instructional detail and support as a learner's proficiency evolves. The approach proposed in this paper aims to address the limitations of existing educational frameworks regarding the specificity and effectiveness of instructional support.

3. Enhancing educational outcomes and overcoming traditional limitations

The challenges highlighted in the state-of-the-art section suggest that future ITS research should focus on developing more flexible and responsive systems that adapt in real time to individual learning progress and needs. Addressing the need for adaptive scaffolding is critical, enabling ITS to dynamically adjust instructional granularity as students' skills evolve. Current systems typically use static granularity, which limits PL. Moreover, prevailing rule-based models fall short in representing the compositional and hierarchical nature of skill acquisition, highlighting the need for models that capture the layered structure of knowledge and skills. There is also a need for expert models capable of operating across varying levels of granularity to improve adaptive scaffolding. Such developments would enhance ITS responsiveness and support more effective PL.

AIA-PAL, a novel approach that integrates agent-based modeling with LLMs within adaptive learning platforms, creates an adaptive framework for tailored educational experiences. It leverages the synergistic strengths of Lang-

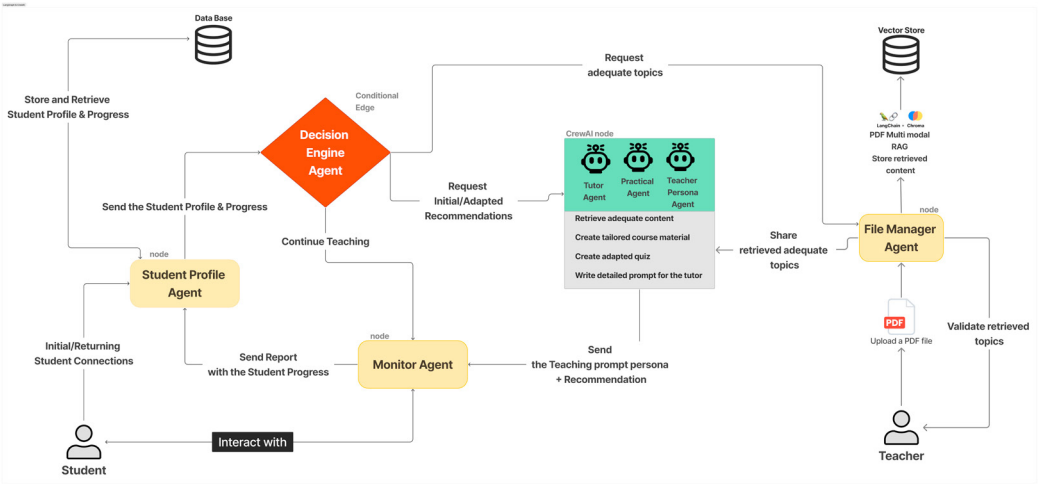

Fig. 1. The AIA-PAL architecture, leveraging LangGraph as the core decision-making engine, and the Education Crew as a single node.

Graph [34, 35] and CrewAI frameworks [36] along with advanced LLMs to develop dynamic, adaptive, and highly PL settings that cater to the unique needs of each student. LangGraph serves as the central decision engine, continuously monitoring student progress and orchestrating adjustments through the Education Crew. The MAS architecture comprises specialized agents that engage students through interactive dialogues, assess their academic ability, and identify areas for improvement. It includes several key agent types: Decision-making agents and Education Crew agents. These agents collaborate to deliver PL paths, comprehensive summaries, and clear explanations, with task complexity dynamically adjusted to match learner progression. The agent-based architecture employs a sophisticated modeling approach where each agent exhibits specific characteristics: **(i) reactive**, continuously monitoring and adjusting to the student's learning progress and immediate needs. **(ii) proactive**, taking the initiative to propose more effective pedagogical strategies. **(iii) social behavior** through inter-agent communication and collaboration.

To ensure the reliability of the content and mitigate the possibility of model hallucination, the system incorporates RAG framework for knowledge management. This technical foundation is complemented by a well-defined teaching methodology based on established pedagogical materials. Human oversight remains integral to the system, and teachers serve as validators who review and approve content generated by agents and LLMs, ensuring alignment with educational standards and maintaining pedagogical quality.

4. Design and architecture of AIA-PAL

Our architecture considers LangGraph as a powerful decision-making engine, while the Education Crew acts as an adaptable agent-based system that dynamically adjusts learning based on LangGraph's insights. Each team ensures that the student's learning experience remains personalized and effective.

4.1. LangGraph as the Core Decision Engine

LangGraph serves as the core decision-making engine, with Education Crew represented as a single node within the LangGraph architecture, as illustrated in Fig. 1. Each agent within this architecture specializes in distinct functions, representing various facets of the learning process. The nodes within the LangGraph architecture are as follows:

- **Student profile Agent (Node)**: It retrieves student activity data from the database, including profile details, skills, preferences, completed modules, quizzes, interactions, and engagement metrics. For new students, it initializes their profile with basic information such as skill level, interests, and learning goals. For returning students, it updates their activity data and sends the profile and progress information to the *Decision Engine Agent* for processing.

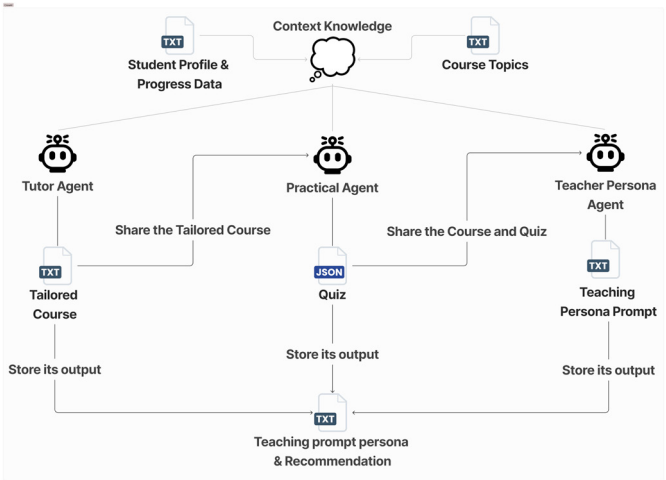

Fig. 2. The Education Crew architecture with CrewAI.

- **Decision Engine Agent (Core Node)**: it analyzes student activity data to determine necessary adjustments to the learning path, quizzes, or teacher persona. For new students, it processes initial profile data to recommend an initial learning path. As students progress, it evaluates their performance, adapting the learning path and quiz content as needed. If a student excels, the system increases challenges; if struggling, it provides additional support, following scaffolding principles. For steady progress, minimal changes are made, reducing assistance over time to promote independence. It directs adjustments to the *Education Crew node* or requests continuation from the *Monitoring Agent node*.
- **Education Crew (Node)**: It comprises various agents—*Tutor, Practical, and Teacher Persona Agents*. It executes initial recommendations and ongoing adaptations based on the Decision Engine Agent's directives and teacher-uploaded PDFs.
- **Monitor Agent (Node)**: It delivers content recommended by the *Education Crew*, adhering to the suggested teacher persona. It interacts with students, monitors engagement, and provides personalized feedback and hints. It tracks responses, including time spent on lessons, quiz performance, and exercise completion. Key metrics, such as lesson completion, quiz results, and signs of disengagement (e.g., prolonged task times or failed quizzes), are reported to the *Student Profile Agent* for further decision-making.
- **File Manager Agent (Node)**: It manages the teacher-provided educational PDFs using the PDF multimodal RAG framework. This process establishes the education agent's knowledge base and ensures content accuracy and pedagogical quality through document preparation, embedding, and storage. Additionally, the File Manager Agent is responsible for coordinating the validation process with the teacher.

4.2. Education Crew as a Node in the LangGraph

The Education Crew, a single node in the global architecture, composed of various educational and pedagogical agents (as illustrated in Fig. 2), is responsible for providing learning recommendations and adaptations. These are informed by the teaching materials, the student profile, and the progress data supplied by the *Decision Making Engine, and File Manager Agents*. The Education Crew performs the following tasks:

- **Create/Adjust Learning Path**: Tutor Agent designs the initial learning path and adjusts it as needed by adding, removing, or updating course content to match the student's progress and pace.
- **Create/Adjust Quizzes and Assessments**: Practical Agent designs initial quizzes/assessments and adjusts them as needed to align with the student's progress, ensuring they match the student's current understanding level.
- **Create/Adjust Teacher Persona**: Teacher Profile Agent creates or adjusts a prompt persona for the Monitor Agent responsible for delivering the recommended course. The agent persona considers the communication

style, student profile, and pedagogical methods. If deeper insights are needed, for example, when a student is struggling with a topic, this agent persona can retrieve external educational resources.

4.3. Implementation of a Dual Approach with CrewAI and LangGraph

In our prototype, we implemented Education Crew as collaborative and adaptive agents using the CrewAI approach, where agents work together to achieve shared goals. Each agent is assigned a specific role and communicates in real time to address strengths and weaknesses. Task execution follows a structured sequential process, ensuring logical progression, while asynchronous execution allows agents to handle urgent tasks outside the main loop. Both processes balance predictable tasks with dynamic real-time interactions, enabling uninterrupted workflows and adaptability. By combining collaboration, adaptability, and dynamic task allocation, the system solves complex problems beyond the capacity of a single agent. The CrewAI Toolkit further enhances the agents capabilities by organizing and storing agent-generated content in a structured format for easy future access. Complementing this adaptive approach, we implemented decision-making agents using the LangGraph framework, which extends LangChain with task-looping functionality for continuous self-questioning and complex actions. LangGraph models programs as stateful graphs, with nodes as computation phases (powered by LLMs) and edges guiding dynamic flows. Nodes act as agents with tools for tasks like database queries, API calls, or engaging external processes via ToolNodes, while edges enable communication, and routers handle dynamic task routing. This structure supports adaptive behavior and efficient problem-solving, making it ideal for real-time decision making. Together, these dual approaches, CrewAI for collaboration and adaptability, and LangGraph for decision making, created a robust system where agents seamlessly coordinated as a team while independently making informed decisions. This enables the system to handle tasks efficiently, adapt to changing conditions, and respond dynamically to both predictable workflows and unforeseen challenges.

5. Development and AIA-PAL's System Evaluation

The proposed educational system is currently under active development. The latest version includes a functional system that demonstrates the key features: **(i) Backend System**: is developed using Python-based MAS components leveraging CrewAI and LangGraph. **(ii) Student profile**: A database is maintained to track individual preferences, progress, and interactions in a given course. **(iii) Frontend System**: Will be built using React, providing an intuitive and interactive user interface. It will display the course content, enable quizzes, and support real-time interactions. The backend is currently operational for core evaluation and testing, with LangSmith [38] tool to track agent workflow. As CrewAI and LangGraph are relatively recent technologies, they are in constant improvement, necessitating regular adaptations to leverage new features and enhancements as they are released.

5.1. In-house evaluation of the agent workflow in AIA-PAL system

To assess and measure the effectiveness of AIA-PAL in providing personalized adaptive learning experiences, we conducted a pilot test as an in-house evaluation of the system. The goal of this pilot test is to assess how well the system functions, particularly in terms of agent workflow efficiency, agent interactions, decision-making, adaptability, and performance. This paper outlines the following key evaluation criteria, built based on our experience in the domain and based on the framework functionalities:

1. **Accuracy** – Do agents provide correct and pedagogically sound responses?
2. **Adaptability** – How well does the system adjust learning paths based on real-time user interactions?
3. **Response Time** – How quickly do agents generate and adjust content?
4. **Workflow Coordination** – Are agents communicating and transitioning tasks effectively?
5. **Track agent decision-making processes** – How many iterations before the right learning path is selected? Do agents iterate effectively to select the optimal learning path?
6. **Errors and inconsistencies** – Does the system provide inappropriate hints? Are responses too slow?
7. **Expert Review and Human-on-the-Loop Evaluation** – Do agents incorporate expert feedback and align decisions with learning objectives?

8. **Scaffolding strategies** – Measuring how well content adjustments align with scaffolding principles, do agents apply scaffolding strategies?

5.2. Preliminary results and discussion

The evaluation was performed using our team, who had prior experience with the AIA-PAL system and had previously interacted with the learning modules in the Python course. This allowed us to assess how well the system adapts to returning users, maintains learning continuity, and refines personalized recommendations over time. The following results summarize our findings based on key evaluation criteria. Additionally, an example of visual representation of the system workflow in RAG trace is provided by LangSmith[1], offering a clear and structured overview of the data flow, key processes, and interactions within the system.

Table 1. Evaluation of the Agent Workflow in AIA-PAL System.

Evaluation Criteria	**Findings**
Accuracy	The RAG Agent ensured content alignment, but the Monitor Agent had minor inconsistencies in explanations.
Adaptability	The Decision Engine Agent adapted learning paths to student progress, and CrewAI adjusted them.
Response Time	RAG Agent: 147.63s (1x), CrewAI: 90.15s. Processing time impacts responsiveness, but is expected in content generation.
Workflow Coordination	Agent coordination was seamless with efficient task transitions, though slight delays occurred in content generation.
Tracking Agents	Agents iterated effectively to determine optimal learning paths. Iterations were minimal.
Errors and Inconsistencies	Occasional infinite loops in the Monitor Agent, but these are easily fixed.
Expert Review	The teacher confirmed the extracted chunks matched the original document.
Scaffolding Principles	The Decision Engine Agent, CrewAI, and Monitor Agent adjusted instructional difficulty dynamically.

6. Conclusion

The AIA-PAL framework represents a significant advancement in personalized adaptive learning by integrating AI-driven technologies, such as LangGraph and CrewAI, within a MAS architecture. By leveraging LLMs and HAI techniques, AIA-PAL dynamically adapts educational content to each learner's needs, overcoming limitations of traditional ITS like static granularity and lack of personalized scaffolding. AIA-PAL ensures both adaptive instruction and teacher oversight, balancing AI-driven content with human validation. The RAG tool enhances reliability by reducing hallucinations in generated content. The pilot test conducted in this paper was built as an in-house test by the developers of the framework to serve as a first evaluation step. Future work will focus on conducting a user evaluation to assess usability, trust, and engagement. The next step of the evaluation will involve participants interacting with AIA-PAL and responding to a structured questionnaire designed to capture both their objective understanding and subjective perception of the system. Evaluation and testing will include user feedback collection through surveys and in-system prompts, as well as LLM performance testing to assess consistency and minimize hallucinations. Addressing challenges such as scalability, reliability, and ethical considerations remains a priority. These improvements will expand AIA-PAL's applicability across diverse subjects and educational needs. Overall, AIA-PAL demonstrates the potential of AI-driven education, paving the way for more personalized, engaging, and effective learning experiences.

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